VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY

HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY

FACULTY OF COMPUTER SCIENCE

_____ * _____

Laboratory Report

LAB 1.3 – Website Layout & Frontend Foundation

Teacher: NGUYỄN ĐỨC THÁI

Class: CC01

—o0o—

Huỳnh Đức Phát (2452940)

Semester project: Medical Clinic

1. Project structure

Wellnest Medical Clinic implements the frontend foundation required by LAB 1.3. In the Medical Clinic domain, catalog items are consultation services. The project is placed under the Apache document root at xamppfiles/htdocs/medical_clinic/.



Figure 1. Actual project directory entries; supporting folders are collapsed.

Required files: public/index.php, public/css/style.css, public/js/script.js, public/images/, pages/item_details.php, components/header.php, and components/footer.php.

PHP require_once includes the shared header and footer on the homepage and additional pages. Supporting app/ and database/ files keep service names, descriptions, categories, images, and prices consistent between pages.

2. Homepage and reusable layout

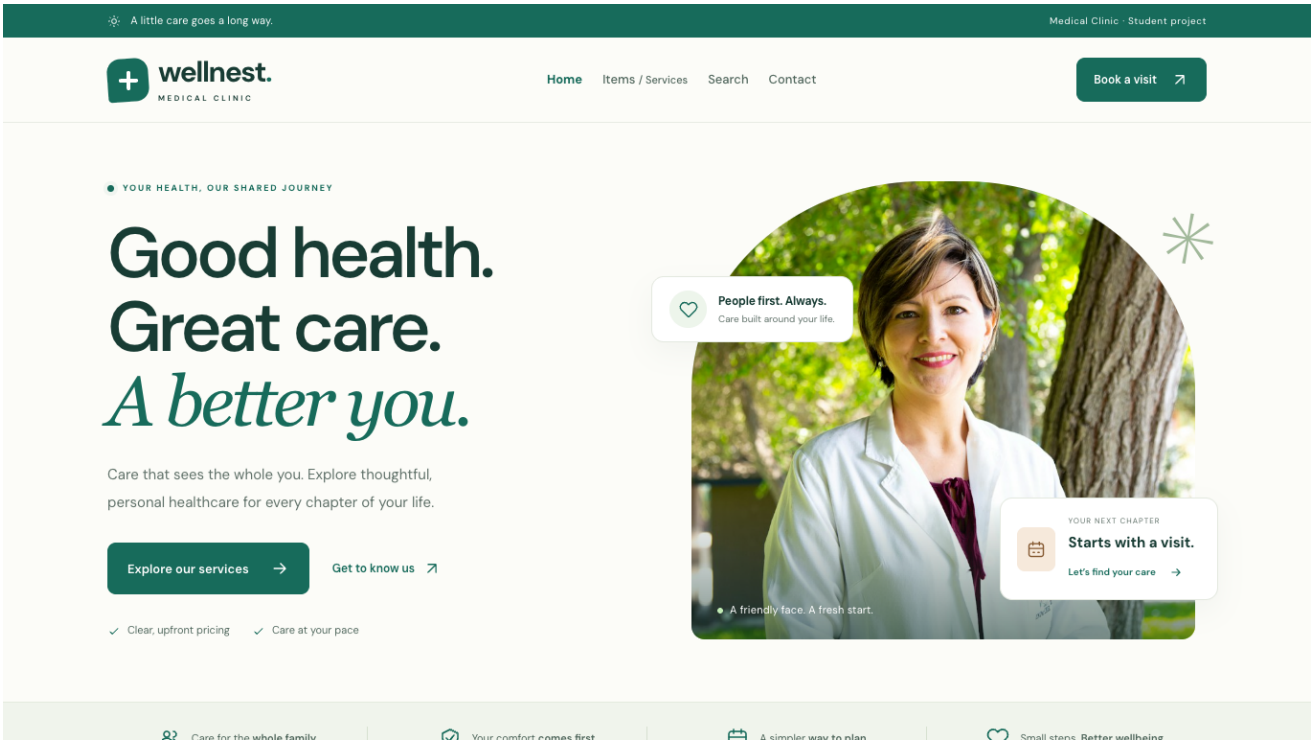


Figure 2. Homepage at the required desktop viewport, 1366 × 768.

The homepage contains a brand header, navigation, hero, item catalog, and footer. The navigation includes Home, Items / Services, Search, and Contact. Book a visit provides a domain-appropriate action. Bootstrap navbar-expand-lg collapses navigation below 992px.

Required feature	Implementation
HTML5 foundation	DOCTYPE, html lang="en", UTF-8, viewport meta, meaningful title; header, nav, main, section, article and footer.
Hero section	Medical Clinic introduction, local doctor photograph, and Explore our services link.
Reusable components	components/header.php and components/footer.php are included using require_once.
Footer	Wellnest name, current-year copyright, illustrative email/location and contact information.
Custom CSS3	Typography, teal/cream palette, spacing, image shapes, cards, buttons, footer, hover effects and responsive media queries.

3. Medical service catalog

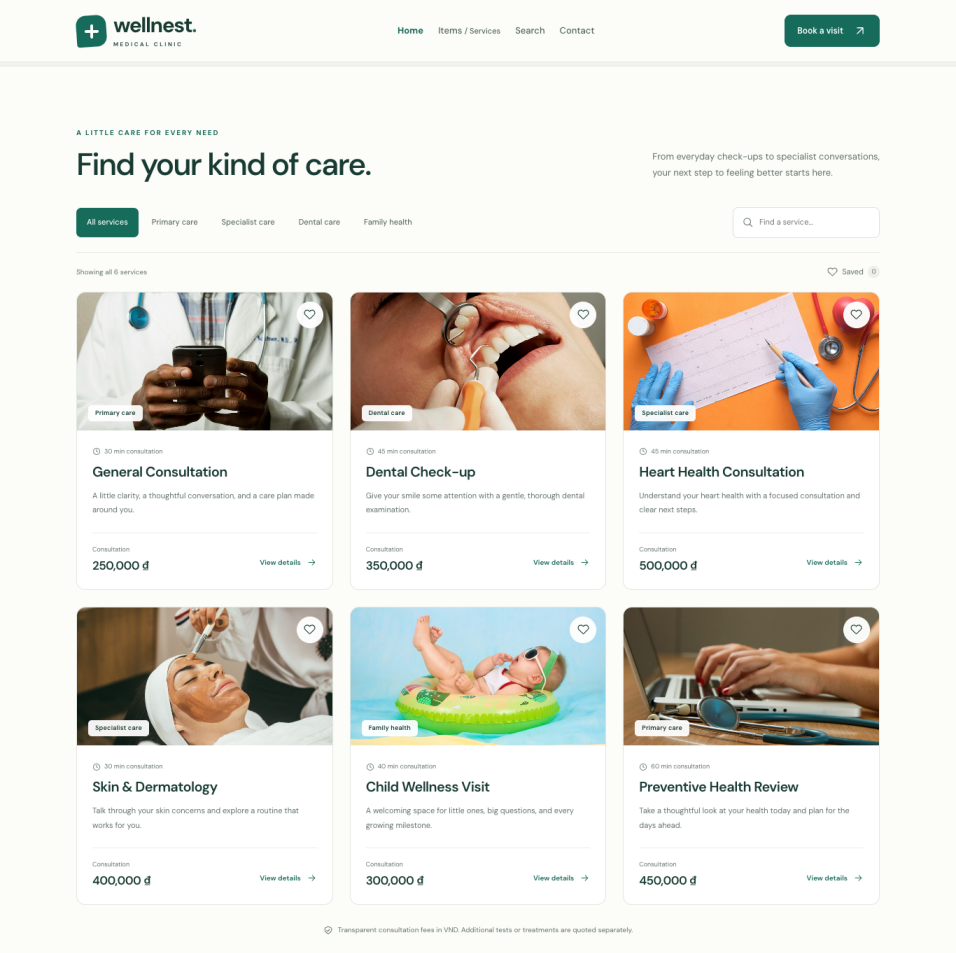


Figure 3. All six service cards in the three-column desktop catalog.

Item name	Category	Price (VND)
General Consultation	Primary care	250,000
Dental Check-up	Dental care	350,000
Heart Health Consultation	Specialist care	500,000
Skin & Dermatology	Specialist care	400,000
Child Wellness Visit	Family health	300,000
Preventive Health Review	Primary care	450,000

Each Bootstrap card contains an image, service name, short description, category, consultation price and View details link. The exact grid classes col-12 col-md-6 col-lg-4 produce one card per row on mobile, two on tablet and three on desktop.

Prices and services are illustrative semester-project content. Stock imagery, brand text, colors, button labels and descriptions are customized for the Medical Clinic domain.

4. Item details page

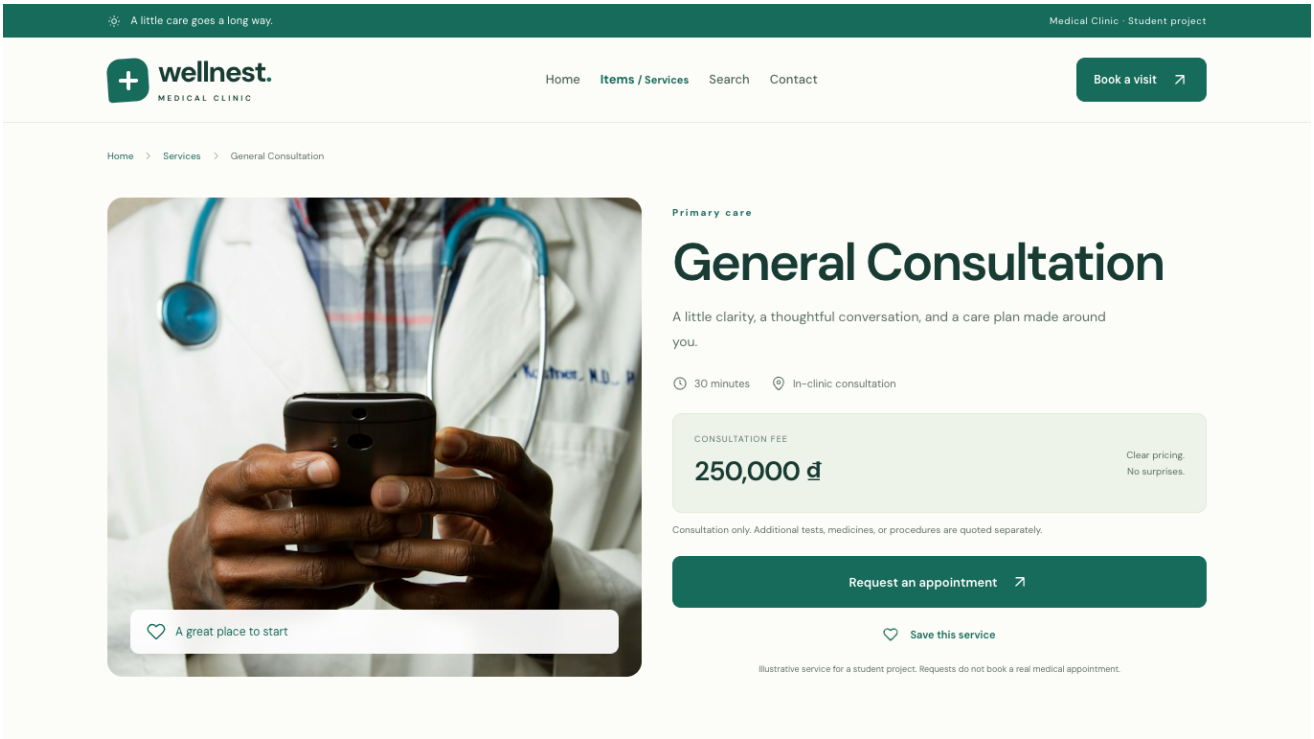


Figure 4. pages/item_details.php?id=1 at 1366 × 768.

Specification requirement	Example on the details page
Item name	General Consultation
Item image	Local healthcare image with descriptive alternative text.
Category	Primary care
Description	Service introduction, full description, inclusions and visit preparation.
Price	250,000 ₫, consistent with the homepage catalog.
Action button	Request an appointment opens the form with this service preselected.

All six item cards open their own details using a validated service ID. The shared header/footer retain consistent navigation and visual styling. An unknown or malformed ID returns an HTTP 404 page with a link back to services.

5. Responsive design: mobile

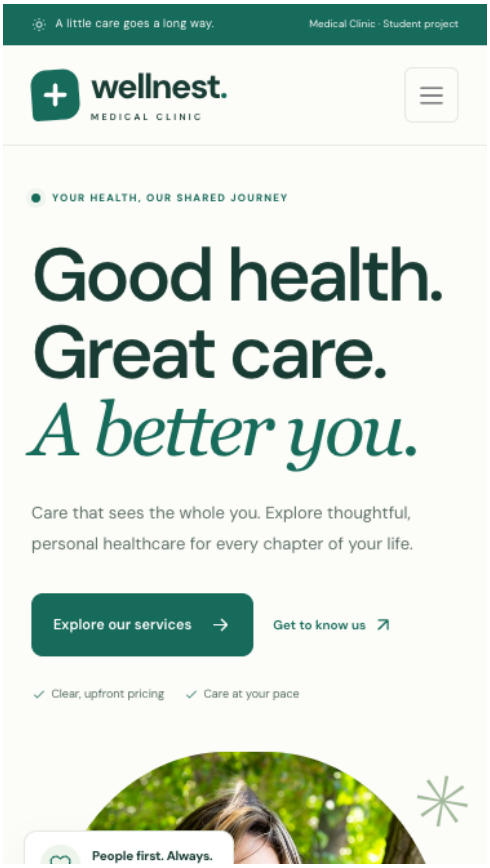


Figure 5. Hero at 375 × 667.

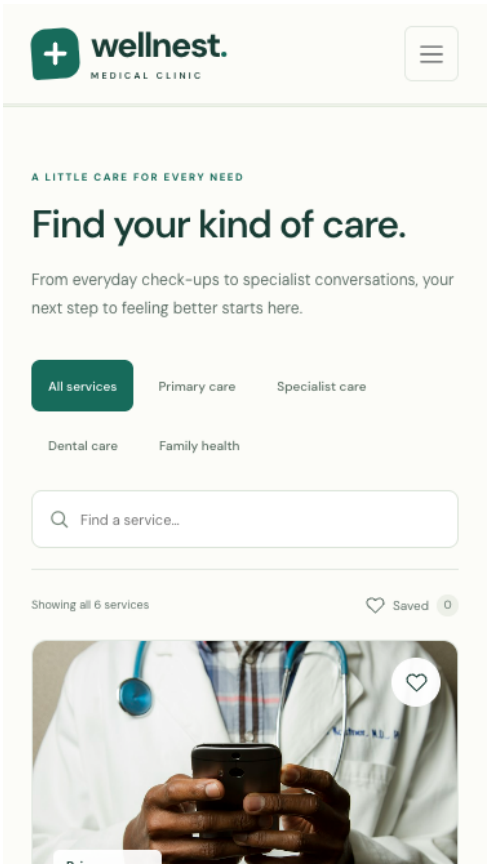


Figure 6. Single-column mobile catalog.

Mobile checks confirm readable text, stacked sections, contained images, accessible controls and no horizontal scrolling. The hamburger opens and closes; selecting Search closes the menu and focuses the service search field.

Required viewport	Catalog layout	Result
Mobile: 375 × 667	1 card per row	Passed
Tablet: 768 × 1024	2 cards per row	Passed
Desktop: 1366 × 768	3 cards per row	Passed

Additional checked widths: 320, 390, 576, 820, 991, 992, 1024 and 1920px. The homepage, service details and appointment form were checked at all 11 sizes. No horizontal overflow or overflowing images was detected.

6. Responsive design: tablet

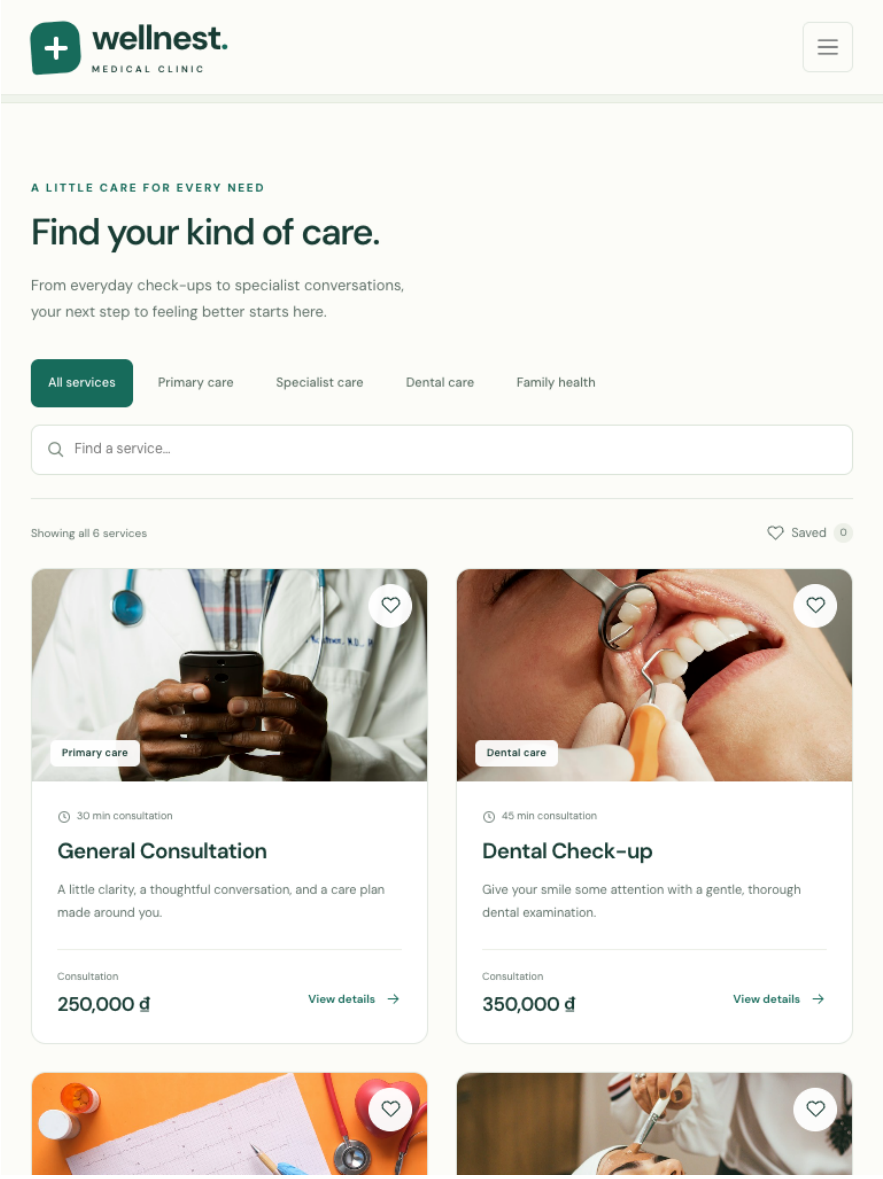


Figure 7. Tablet catalog at the required 768 × 1024 viewport.

At the md breakpoint (768px), the service cards form two columns. The search field and category controls adapt to the available width. Navigation continues to use the collapsible menu until the lg breakpoint (992px).

At desktop widths the catalog forms three columns and the navigation expands. Local images use constrained widths, object-fit and responsive containers. The detail page changes from two side-by-side columns to a stacked layout on smaller screens.

7. JavaScript interaction

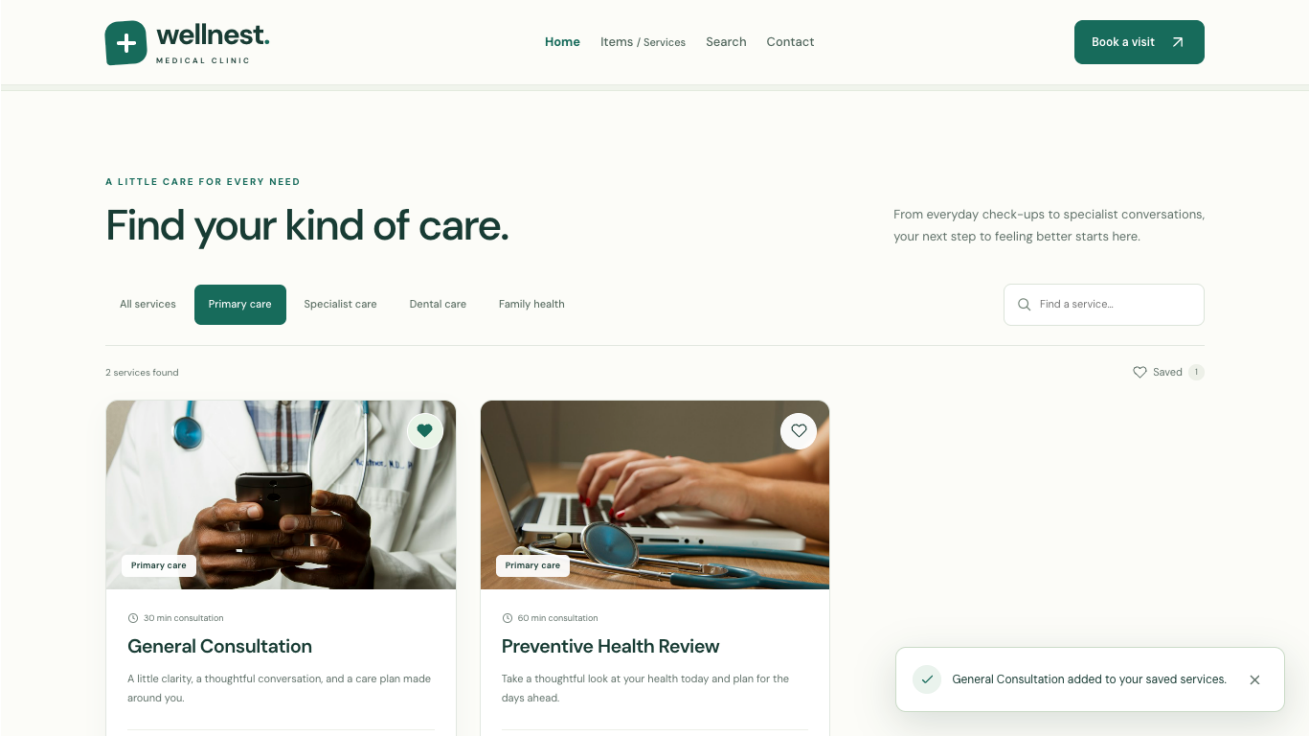


Figure 8. Primary care filter, saved service state and visible confirmation toast.

The required visible JavaScript interaction is demonstrated by saving General Consultation: its heart button changes state, the saved count updates, and a notification announces the action. Saved selections persist in localStorage and remain available after refresh.

Feature	Demonstration
Challenge A: Search	Type preventive to show the matching service; a query without matches shows a resettable empty state.
Challenge B: Categories	Choose Primary care to display only General Consultation and Preventive Health Review. Search and category filters combine.
Challenge E: Interactive cards	Save or unsave a service, filter saved items, refresh to verify persistence, or open its details.
Additional form interaction	Choose a service to update its displayed fee; submit a valid demo request to see a reference and confirmation. The interface explicitly identifies this as a demo.

Demonstration steps: open Items / Services; select Primary care; click the General Consultation heart; observe the changed heart, saved count and toast. These actions are implemented in public/js/script.js.

8. Accessibility and completion

Basic accessibility checks cover meaningful page titles, descriptive image alt attributes, a logical h1–h2–h3 hierarchy, meaningful link labels, text/background contrast, visible keyboard focus, a skip link, form labels and non-image text for important information. Toggle states use aria-pressed/aria-expanded; search results and notifications use live regions.

Automated axe-core WCAG A/AA checks were run on the homepage, details and form at 375px and 1366px: six scans completed with zero reported violations. Keyboard navigation and the mobile menu were also exercised. Reduced-motion preferences disable nonessential transitions.

Lab completion checklist	Status
Project folder under htdocs and required folder structure	Completed
public/index.php and semantic HTML5 structure	Completed
Bootstrap added with a local fallback	Completed
Responsive navigation and hero section	Completed
Six service cards and responsive 1 / 2 / 3-column grid	Completed
Custom CSS with more than three visual customizations	Completed
Reusable header and footer components	Completed
Item details: name, image, category, description, price, action	Completed
Working JavaScript with visible interaction feedback	Completed
Mobile, tablet and desktop layouts tested	Completed
Meaningful alt text, titles, headings, links and contrast checked	Completed
Medical Clinic content replaces generic sample content	Completed
All required screenshots captured	Completed

Submission evidence

The complete project folder accompanies this report. Required screenshots are in report/screenshots/: 01-project-structure.png; 02-homepage-desktop.png; 03-homepage-mobile.png; 05-item-details.png; 06-javascript-interaction.png. Additional full-page, tablet, navigation and form screenshots are included. Browser verification results are stored in report/verification.json.